



**Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
*Institut Teknologi Sepuluh Nopember***

Laporan Akhir Praktikum Jaringan Komputer

Vlan-Trunk-OSPF-MultiVendor

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1 Langkah-Langkah Percobaan

1.1 Praktikum 1 Konfigurasi Cisco Switch

1. Masuk ke mode konfigurasi pada Cisco Switch menggunakan perintah enable dan configure terminal.
2. Membuat VLAN 10, VLAN 20, VLAN 30, dan VLAN 40 sesuai dengan kebutuhan jaringan.
3. Melakukan verifikasi VLAN menggunakan perintah show vlan brief untuk memastikan seluruh VLAN telah berhasil dibuat dan berstatus aktif.

```
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 30
Switch(config-vlan)#
Switch(config-vlan)#name IT
Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 40
Switch(config-vlan)#
Switch(config-vlan)#name MARKETING
Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#exit
Switch#
Switch#show
*Jun  5 02:16:22.946: N5YS-S-CONFIG_I: Configured from console by consolevlan br
VLAN Name                Status    Ports
-----
1    default                active    Gi0/8, Gi0/1, Gi0/2, Gi0/3
20   FINANCE                 active    Gi1/0, Gi1/1, Gi1/2, Gi1/3
30   IT                      active
40   MARKETING               active
1002 fddi-default            act/unsup
1003 token-ring-default     act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup
Switch#
Switch#
```

Gambar 1

4. Mengonfigurasi port yang terhubung ke masing-masing PC sebagai access port dan menempatkannya pada VLAN yang sesuai.
5. Mengonfigurasi interface yang terhubung ke Cisco Router sebagai trunk port agar dapat membawa VLAN 10, 20, 30, dan 40.
6. Melakukan verifikasi konfigurasi trunk menggunakan perintah show interfaces trunk.

```
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#
Switch(config-if)#description TRUNK-TO-CISCO-ROUTER
Switch(config-if)#
Switch(config-if)#switchport mode trunk
Switch(config-if)#
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40
Switch(config-if)#
Switch(config-if)#no shutdown
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#
Switch#
Switch#show
*Jun  5 02:22:07.368: N5YS-S-CONFIG_I: Configured from console by console
A Type "show ?" for a list of subcommands
Switch#
Switch#show interfaces trunk
Port      Mode          Encapsulation  Status        Native vlan
-----
Gi0/0     on            802.1q         trunking      1
Gi0/0     Vlans allowed on trunk
          10,20,30,40
Gi0/0     Vlans allowed and active in management domain
          10,20,30,40
Gi0/0     Vlans in spanning tree forwarding state and not pruned
          10,20,30,40
Switch#
Switch#
```

Gambar 2

7. Menyimpan konfigurasi menggunakan perintah write memory.

1.2 Praktikum 2 Konfigurasi Cisco Router

1. Masuk ke mode konfigurasi Cisco Router.
2. Mengonfigurasi interface yang terhubung ke Cisco Switch sebagai trunk interface.

3. Membuat subinterface untuk VLAN 10, VLAN 20, VLAN 30, dan VLAN 40 menggunakan metode Router on a Stick (ROAS).
4. Memberikan alamat IP gateway pada masing-masing subinterface VLAN.
5. Mengonfigurasi DHCP Server untuk VLAN 10 dan VLAN 20.
6. Mengonfigurasi alamat IP statis pada PC VLAN 30 dan VLAN 40.
7. Melakukan verifikasi interface menggunakan perintah show ip interface brief.

```

Jun  5 02:31:51.489: MGRUB-5-CONFIG-WRITTEN: GRUB configuration was written to disk successfully.
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#exit
Router#
Router#
Jun  5 02:32:01.014: NSYS-5-CONFIG-I: Configured from console by console
Router#
Router#write memory
Building configuration...
[OK]
Router#
Jun  5 02:32:12.963: MGRUB-5-CONFIG-WRITING: GRUB configuration is being updated on disk. Please wait...
Jun  5 02:32:13.649: MGRUB-5-CONFIG-WRITTEN: GRUB configuration was written to disk successfully.
Router#
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0      unassigned      YES unset    administratively down down
GigabitEthernet0/1      unassigned      YES unset    administratively down down
GigabitEthernet0/2      unassigned      YES unset    up
GigabitEthernet0/2.10   192.168.10.1    YES manual   up
GigabitEthernet0/2.20   192.168.20.1    YES manual   up
GigabitEthernet0/2.30   192.168.30.1    YES manual   up
GigabitEthernet0/2.40   192.168.40.1    YES manual   up
GigabitEthernet0/3      unassigned      YES unset    administratively down down
Router#
Router#

```

Gambar 3

8. Melakukan verifikasi DHCP Pool menggunakan perintah show ip dhcp pool.

```

Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0      unassigned      YES unset    administratively down down
GigabitEthernet0/1      unassigned      YES unset    administratively down down
GigabitEthernet0/2      unassigned      YES unset    up
GigabitEthernet0/2.10   192.168.10.1    YES manual   up
GigabitEthernet0/2.20   192.168.20.1    YES manual   up
GigabitEthernet0/2.30   192.168.30.1    YES manual   up
GigabitEthernet0/2.40   192.168.40.1    YES manual   up
GigabitEthernet0/3      unassigned      YES unset    administratively down down
Router#
Router#show ip dhcp pool
Pool VLAN10-FINANCE :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index  IP address range      Leased addresses
    192.168.10.12  192.168.10.1 - 192.168.10.254  1
Pool VLAN20-HR :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index  IP address range      Leased addresses
    192.168.20.12  192.168.20.1 - 192.168.20.254  1
Router#
Router#

```

Gambar 4

9. Melakukan verifikasi DHCP Binding menggunakan perintah show ip dhcp binding.

```

GigabitEthernet0/3      unassigned      YES unset    administratively down down
Router#
Router#show ip dhcp pool
Pool VLAN10-FINANCE :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index  IP address range      Leased addresses
    192.168.10.12  192.168.10.1 - 192.168.10.254  1
Pool VLAN20-HR :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index  IP address range      Leased addresses
    192.168.20.12  192.168.20.1 - 192.168.20.254  1
Router#
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6666.41 Jun 06 2026 02:35 AM  Automatic
192.168.20.11    0100.5079.6666.40 Jun 06 2026 02:30 AM  Automatic
Router#
Router#

```

Gambar 5

10. Melakukan pengujian DHCP pada PC VLAN 10, VLAN 20, VLAN 30 dan VLAN 40.

```

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Dailing
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS>
VPCS> ip dhcp
DHCP: IP 192.168.10.11/24 GW 192.168.10.1

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER: 192.168.10.1
DHCP LEASE : 86142, 86400/43200/75600
MAC        : 08:50:79:66:68:41
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS>
```

Gambar 6: PC VLAN 10

```

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Press '?' to get help.

VPCS>
VPCS> ip dhcp
DHCP: IP 192.168.10.11/24 GW 192.168.10.1

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER: 192.168.10.1
DHCP LEASE : 86142, 86400/43200/75600
MAC        : 08:50:79:66:68:41
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.10.1

64 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=2.770 ms
64 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=1.696 ms
64 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=1.437 ms
64 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=1.834 ms
64 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=2.144 ms

VPCS>
```

Gambar 7: Ping

```

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Press '?' to get help.

VPCS>
VPCS> ip dhcp
DHCP: IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER: 192.168.20.1
DHCP LEASE : 86400, 86400/43200/75600
MAC        : 08:50:79:66:68:40
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.20.1

64 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=1.519 ms
64 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=1.774 ms
64 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=1.447 ms
64 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=1.513 ms
64 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=1.712 ms

VPCS>
```

Gambar 8: PC VLAN 20

```

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Press '?' to get help.

VPCS>
VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.30.10/24
GATEWAY    : 192.168.30.1
DNS        :
MAC        : 08:50:79:66:68:3f
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.30.1

64 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=1.825 ms
64 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=1.485 ms
64 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=1.899 ms
64 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=1.746 ms
64 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=1.979 ms

VPCS>
```

Gambar 9: PC VLAN 30

```

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Press '?' to get help.

VPCS>
VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
*HPC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.40.10/24
GATEWAY    : 192.168.40.1
DNS        :
MAC        : 08:50:79:66:60:3e
LPORT     : 20000
RHOST-PORT : 227.6.0.1:30000
MTU        : 1500

VPCS> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=1.617 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=1.731 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=2.004 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=1.394 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=1.175 ms

VPCS>

```

Gambar 10: PC VLAN 40

11. Uji ping antar-Vlan dari vlan 10

```

VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=2.770 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=1.696 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=1.637 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=1.834 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=2.104 ms

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=3.717 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=3.631 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=3.539 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=3.868 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=3.915 ms

VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=1.927 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=1.130 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=2.899 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=3.663 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=3.067 ms

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=2.632 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.809 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=2.248 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=3.630 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=5.395 ms

VPCS>

```

Gambar 11

12. Masuk ke cisco router lagi untuk melihat ip dhcp binding

```

Leased addresses      : 1
Pending event         : none
1 subnet is currently in the pool :
Current index  IP address range  Leased addresses
192.168.10.12    192.168.10.1 - 192.168.10.254    1

Pool VLAN20-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses              : 254
Leased addresses             : 1
Pending event                : none
1 subnet is currently in the pool :
Current index  IP address range  Leased addresses
192.168.20.12  192.168.20.1 - 192.168.20.254    1

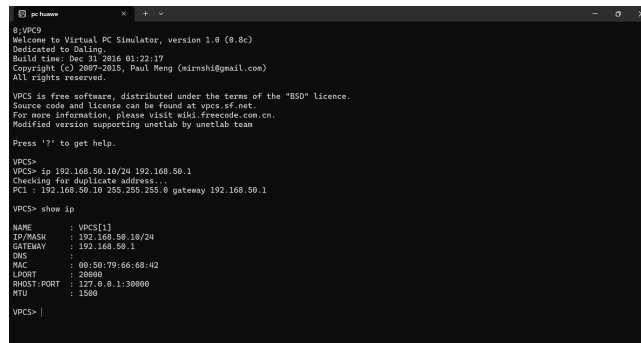
Router#
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668-41 Jun 06 2026 02:35 AM    Automatic
192.168.20.11    0100.5079.6668-40 Jun 06 2026 02:34 AM    Automatic
Router#
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668-41 Jun 06 2026 02:35 AM    Automatic
192.168.20.11    0100.5079.6668-40 Jun 06 2026 02:34 AM    Automatic
Router#
Router#

```

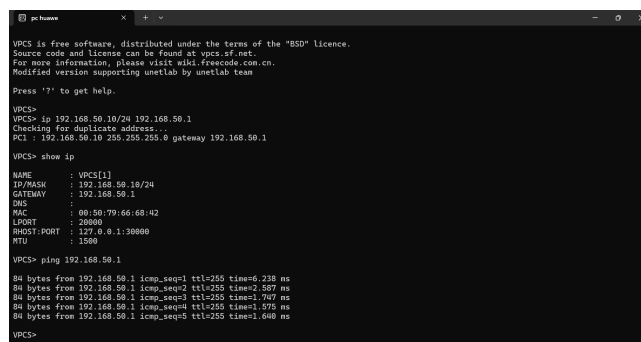
Gambar 12

1.3 Praktikum 3 Konfigurasi Huawei Router dan IP Address Antar Router

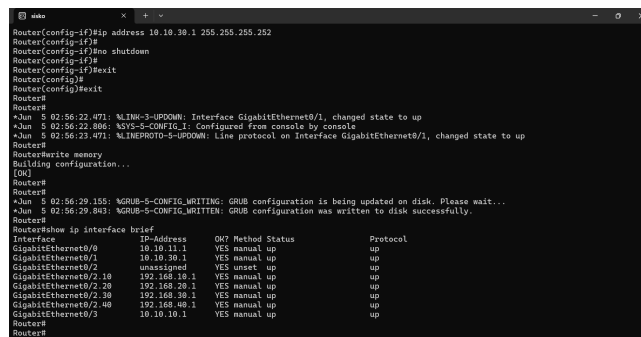
1. Mengonfigurasi interface LAN pada Huawei Router dengan jaringan 192.168.50.0/24.



2. Mengonfigurasi alamat IP pada PC yang terhubung ke LAN Huawei.
3. Melakukan pengujian konektivitas antara PC LAN Huawei dengan gateway.



4. Mengonfigurasi link pertama antara Cisco Router dan Huawei Router.
5. Mengonfigurasi link kedua antara Cisco Router dan Huawei Router.
6. Mengonfigurasi link antara Huawei Router dan MikroTik.
7. Mengonfigurasi link antara Cisco Router dan MikroTik.
8. Melakukan verifikasi interface pada Cisco Router menggunakan perintah show ip interface brief.



9. Melakukan verifikasi interface pada Huawei Router menggunakan perintah display ip interface brief.


```

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=1 ms

--- 10.10.11.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

--- 10.10.20.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/1 ms

[~HUAWEI]
[~HUAWEI]

```

Gambar 20: Dari Router Huawei(2)

```

Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 10.10.20.2/30 10.10.20.0 ether1
1 10.10.30.2/30 10.10.30.0 ether2
[admin@Mikrotik] > ping 10.10.20.1
seq=0 ttl=64 size=56 time=0.255 ms
0 10.10.20.1 56 255 1ms
1 10.10.20.1 56 255 0ms
2 10.10.20.1 56 255 0ms
3 10.10.20.1 56 255 1ms
4 10.10.20.1 56 255 0ms
5 10.10.20.1 56 255 0ms
6 10.10.20.1 56 255 0ms
7 10.10.20.1 56 255 0ms
8 10.10.20.1 56 255 0ms
9 10.10.20.1 56 255 0ms
10 10.10.20.1 56 255 0ms
sent=11 received=11 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=1ms

[admin@Mikrotik] > ping 10.10.30.1
seq=0 ttl=64 size=56 time=0.255 ms
0 10.10.30.1 56 255 0ms
1 10.10.30.1 56 255 1ms
2 10.10.30.1 56 255 1ms
3 10.10.30.1 56 255 0ms
4 10.10.30.1 56 255 1ms
5 10.10.30.1 56 255 2ms
6 10.10.30.1 56 255 0ms
sent=7 received=7 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=2ms

[admin@Mikrotik] >

```

Gambar 21: Dari Router Mikrotik

1.4 Praktikum 4 Konfigurasi OSPF Multivendor dan OSPF Cost

1. Mengonfigurasi OSPF pada Cisco Router dan mengiklankan seluruh network yang dimiliki.

```

Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Sending updates every 30 seconds
  Invalid after 3 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 16
  Routing for Networks:
    Routing Information Sources:
      Gateway         Distance      Last Update
    Distance: (default is 0)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 16
  Routing for Networks:
    10.10.0.0/24 area 0
    10.10.10.0/24 area 0
    10.10.30.0/24 area 0
    192.168.10.0/24 area 0
    192.168.20.0/24 area 0
    192.168.30.0/24 area 0
    192.168.40.0/24 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    Distance: (default is 110)

Router#

```

Gambar 22

2. Mengatur nilai OSPF Cost pada masing-masing interface Cisco Router.
3. Mengonfigurasi OSPF pada Huawei Router dan mengiklankan seluruh network yang dimiliki.
4. Mengatur nilai OSPF Cost pada interface Huawei Router.
5. Mengonfigurasi OSPF pada MikroTik serta mengatur cost sebagai jalur cadangan.
6. Melakukan verifikasi neighbor OSPF pada Cisco Router.

```

Distance: (default is 0)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)

Router#
*Jun  5 03:20:38.246: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOADING to FULL, Loading Done
*Jun  5 03:20:40.569: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
Router#
*Jun  5 03:27:32.478: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
*Jun  5 03:28:01.463: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
Router#show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1         1   FULL/BDN       00:00:36    10.10.30.2     GigabitEthernet0/1
2.2.2.2         1   FULL/BDN       00:00:36    10.10.11.2     GigabitEthernet0/0
3.3.3.3         1   FULL/BDN       00:00:29    10.10.10.2     GigabitEthernet0/3

```

Gambar 23

7. Melakukan verifikasi OSPF pada Huawei Router.

```

[~HUAWEI]
[~HUAWEI]
[~HUAWEI]display ospf routing

OSPF Process 1 with Router ID 2.2.2.2
  Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
10.10.10.0/24   10        Direct    10.10.10.2    2.2.2.2        0.0.0.0
10.10.11.0/24   10        Direct    10.10.11.2    2.2.2.2        0.0.0.0
10.10.20.0/24   100       Direct    10.10.20.1    2.2.2.2        0.0.0.0
10.10.30.0/24   110       Stub      10.10.11.1    1.1.1.1        0.0.0.0
10.10.30.0/24   110       Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.10.0/24 11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.10.0/24 11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.20.0/24 11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.20.0/24 11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.30.0/24 11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.30.0/24 11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.40.0/24 11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.40.0/24 11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.50.0/24 1        Direct    192.168.50.1  2.2.2.2        0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

---- More ----

```

Gambar 24

8. Melakukan verifikasi OSPF Neighbor pada MikroTik.

```

[admin@Mikrotik] > ping 10.10.30.1
SEQ HOST                      SIZE TTL TIME STATUS
0 10.10.30.1                  56 255 0ms 56 255 0ms
1 10.10.30.1                  56 255 1ms 56 255 1ms
2 10.10.30.1                  56 255 1ms 56 255 1ms
3 10.10.30.1                  56 255 0ms 56 255 0ms
4 10.10.30.1                  56 255 1ms 56 255 1ms
5 10.10.30.1                  56 255 1ms 56 255 1ms
6 10.10.30.1                  56 255 0ms 56 255 0ms

[admin@Mikrotik] > /routing ospf instance set default router-id=3.3.3.3
[admin@Mikrotik] > /routing ospf
[admin@Mikrotik] /routing ospf network add network=10.10.20.0/24 area=backbone
[admin@Mikrotik] /routing ospf network add network=10.10.30.0/24 area=backbone
[admin@Mikrotik] > /routing ospf interface
[admin@Mikrotik] /routing ospf interface add interface=ether1 cost=100
[admin@Mikrotik] /routing ospf interface add interface=ether2 cost=100
[admin@Mikrotik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state=Full state-changes=5 ls-retransmits=0 ls-requests=0 db-summary=0
  adjacency=12s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=10.10.20.2 state=Full state-changes=5 ls-retransmits=0 ls-requests=0 db-summary=0
  adjacency=12s
[admin@Mikrotik] >

```

Gambar 25

```

[~HUAWEI]
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
  Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
State: Full           Mode:Nbr is Slave      Priority: 1
DR: 10.10.10.1        BDR: 10.10.10.2      MTU: 1500
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00h00m38s
Neighbor Up Time : 2026-06-05 03:20:50
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
State: Full           Mode:Nbr is Slave      Priority: 1
DR: 10.10.11.1        BDR: 10.10.11.2      MTU: 1500
Dead timer due in 36 sec
Retrans timer interval: 5
Neighbor is up for 00h00m31s
Neighbor Up Time : 2026-06-05 03:20:50
Authentication Sequence: [ 0 ]

[~HUAWEI]

```

Gambar 26: Display OSPF Peer

```

smt-7 received7 packet-loss0% size-rtt-0ms avg-rtt-0ms max-rtt-2ms

[admin@Mikrotik] / > /routing ospf instance set default router-id 3.3.3.3
[admin@Mikrotik] / > routing ospf
[admin@Mikrotik] /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@Mikrotik] /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@Mikrotik] /routing ospf /
[admin@Mikrotik] / > routing ospf interface
[admin@Mikrotik] /routing ospf interface add interface=ether1 cost=100
[admin@Mikrotik] /routing ospf interface add interface=ether2 cost=100
[admin@Mikrotik] /routing ospf interface /
[admin@Mikrotik] / > /routing ospf neighbor print
0 instance=default router-id=3.3.3.3 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state=Full state-changes=5 ls-retransmits=0 ls-requests=0 db-summary=0
  adjacency=123
1 instance=default router-id=3.3.3.3 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=10.10.20.2 state=Full state-changes=5 ls-retransmits=0 ls-requests=0 db-summary=0
  adjacency=123
[admin@Mikrotik] / > /ip route print show-ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
  6 - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PGPR-SEC      GATEWAY      DISTANCE
0 Ado 10.10.10.0/30  0 10.10.20.1  110
1 Ado 10.10.11.0/30  0 10.10.20.1  110
2 Ado 192.168.10.0/24  0 10.10.30.1  110
3 Ado 192.168.20.0/24  0 10.10.30.1  110
4 Ado 192.168.30.0/24  0 10.10.30.1  110
5 Ado 192.168.40.0/24  0 10.10.30.1  110
6 Ado 192.168.50.0/24  0 10.10.30.1  110
[admin@Mikrotik] / >

```

Gambar 29: Routing Table di Mikrotik

9. Melakukan verifikasi routing table pada seluruh router.

```

10.10.10.0 0.0.0.3 area 0
10.10.11.0 0.0.0.3 area 0
10.10.30.0 0.0.0.3 area 0
192.168.10.0 0.0.0.255 area 0
192.168.20.0 0.0.0.255 area 0
192.168.30.0 0.0.0.255 area 0
192.168.40.0 0.0.0.255 area 0
192.168.50.0 0.0.0.255 area 0

Routing Information Sources:
Gateway      Distance      Last Update
Distance: (default is 110)

Router#
*Jun  5 03:28:38.246: NSPFF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOADING to FULL, Loading Done
*Jun  5 03:28:40.049: NSPFF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOADING to FULL, Loading Done
Router#
*Jun  5 03:27:38.478: NSPFF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
*Jun  5 03:28:01.463: NSPFF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
Router# show ip ospf neighbor

Neighbor ID  Pri  State           Dead Time   Address        Interface
2.2.2.2      1    FULL/DR         00:00:36    10.10.11.2     GigabitEthernet0/3
2.2.2.2      1    FULL/DR         00:00:29    10.10.10.2     GigabitEthernet0/3

Router# show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/3, 00:09:34 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:09:34 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
    10.10.10.2, from 2.2.2.2, 00:09:04 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
  Router#

```

Gambar 27

```

public_Routing Table : OSPF
  Destinations : 9      Routes : 14

OSPF routing table status : <Active>
  Destinations : 5      Routes : 10

Destination/Mask  Proto  Pre  Cost      Flags NextHop      Interface
10.10.30.0/30     OSPF   10   110        0 10.10.11.1     Ethernet1/0/0/4
10.10.11.0/30     OSPF   10   110        0 10.10.10.1     Ethernet1/0/0/4
192.168.10.0/24   OSPF   10   11         0 10.10.11.1     Ethernet1/0/0/4
192.168.20.0/24   OSPF   10   11         0 10.10.11.1     Ethernet1/0/0/4
192.168.30.0/24   OSPF   10   11         0 10.10.10.1     Ethernet1/0/0/4
192.168.40.0/24   OSPF   10   11         0 10.10.11.1     Ethernet1/0/0/4
192.168.50.0/24   OSPF   10   11         0 10.10.10.1     Ethernet1/0/0/4

OSPF routing table status : <Inactive>
  Destinations : 4      Routes : 4

Destination/Mask  Proto  Pre  Cost      Flags NextHop      Interface
10.10.10.0/30     OSPF   10   10         0 10.10.10.2     Ethernet1/0/0/4
10.10.11.0/30     OSPF   10   10         0 10.10.11.2     Ethernet1/0/0/4
10.10.30.0/30     OSPF   10   100        0 10.10.20.3     Ethernet1/0/0/2
192.168.50.0/24   OSPF   10   1          0 192.168.50.1   Ethernet1/0/0/2

[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]

```

Gambar 28: Routing Table di Huawei

10. Melakukan pengujian konektivitas end-to-end antara VLAN Cisco dan LAN Huawei.

```

VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=3.717 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=3.631 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=3.539 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=3.068 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=2.915 ms

VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=1.927 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=2.120 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=2.899 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=3.603 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=3.607 ms

VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=2.622 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.809 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=2.148 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=3.638 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=3.395 ms

VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.538 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.052 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.165 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.022 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.306 ms

VPCS>

```

Gambar 30: dari VPC10

```
VRCS> ip dhcp
DOORA IP 192.168.20.11/24 GW 192.168.20.1

VRCS> show ip

NAME      : VPCS[1]
IP/MASK   : 192.168.20.11/24
GATEWAY   : 192.168.20.1
DNS       : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE : 86400, 86400/43200/75600
MAC       : 08:50:79:66:68:40
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU       : 1500

VRCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=1.519 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=1.776 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=1.417 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=1.513 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=1.712 ms

VRCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.207 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.655 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.313 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.561 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.373 ms

VRCS>
```

Gambar 31: dari VPC20

```
VRCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address
PCI : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VRCS> show ip

NAME      : VPCS[1]
IP/MASK   : 192.168.30.10/24
GATEWAY   : 192.168.30.1
DNS       :
MAC       : 08:50:79:66:68:3f
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU       : 1500

VRCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=1.825 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=1.485 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=1.899 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=1.746 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=1.979 ms

VRCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.258 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.974 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=2.944 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.813 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.905 ms

VRCS>
```

Gambar 32: dari VPC30

```
VRCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
*HPC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VRCS> show ip

NAME      : VPCS[1]
IP/MASK   : 192.168.40.10/24
GATEWAY   : 192.168.40.1
DNS       :
MAC       : 08:50:79:66:68:3e
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU       : 1500

VRCS> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=1.637 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=1.731 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=2.044 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=1.394 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=1.175 ms

VRCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=2.846 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.865 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=4.540 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.746 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=4.141 ms

VRCS>
```

Gambar 33: dari VPC40

```
pc huawei> ping 192.168.10.11

84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=3.906 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=3.848 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=3.975 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=3.681 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=4.118 ms

VRCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=4.620 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=4.977 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=4.534 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=3.633 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=62 time=3.803 ms

VRCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=3.231 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=4.432 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=4.670 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=3.877 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=4.630 ms

VRCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=4.208 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=2.802 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=3.711 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=6.683 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=4.352 ms

VRCS>
```

Gambar 34: dari pc huawei

1.5 Praktikum 5 Pengujian Failover OSPF

1. Melakukan pengujian routing pada kondisi normal saat kedua link Cisco-Huawei aktif.

```
NAME      : VPCS[1]
IP/MASK   : 192.168.38.18/24
GATEWAY   : 192.168.38.1
DNS       :
MAC       : 08:50:79:66:68:3f
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU       : 1500

VPCS> ping 192.168.38.1

84 bytes from 192.168.38.1 icmp_seq=1 ttl=255 time=1.825 ms
84 bytes from 192.168.38.1 icmp_seq=2 ttl=255 time=1.485 ms
84 bytes from 192.168.38.1 icmp_seq=3 ttl=255 time=1.899 ms
84 bytes from 192.168.38.1 icmp_seq=4 ttl=255 time=1.746 ms
84 bytes from 192.168.38.1 icmp_seq=5 ttl=255 time=1.979 ms

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.258 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.974 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=2.941 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.813 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.905 ms

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.38.1  4.383 ms  2.333 ms  2.456 ms
 2 18.18.11.2  4.307 ms  2.418 ms  3.309 ms
 3 *192.168.50.10  3.744 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS>
```

Gambar 35

2. Menonaktifkan salah satu link Cisco-Huawei dan melakukan pengujian kembali.

```
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU       : 1500

VPCS> ping 192.168.38.1

84 bytes from 192.168.38.1 icmp_seq=1 ttl=255 time=1.825 ms
84 bytes from 192.168.38.1 icmp_seq=2 ttl=255 time=1.485 ms
84 bytes from 192.168.38.1 icmp_seq=3 ttl=255 time=1.899 ms
84 bytes from 192.168.38.1 icmp_seq=4 ttl=255 time=1.746 ms
84 bytes from 192.168.38.1 icmp_seq=5 ttl=255 time=1.979 ms

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.258 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.974 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=2.941 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.813 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.905 ms

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.38.1  4.383 ms  2.333 ms  2.456 ms
 2 18.18.11.2  4.307 ms  2.418 ms  3.309 ms
 3 *192.168.50.10  3.744 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.38.1  1.928 ms  1.638 ms  3.372 ms
 2 18.18.11.2  2.728 ms  3.551 ms  2.538 ms
 3 *192.168.50.10  5.579 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS>
```

Gambar 36

3. Menonaktifkan kedua link Cisco-Huawei dan mengamati perpindahan jalur routing ke MikroTik.

```
84 bytes from 192.168.38.1 icmp_seq=2 ttl=255 time=1.485 ms
84 bytes from 192.168.38.1 icmp_seq=3 ttl=255 time=1.899 ms
84 bytes from 192.168.38.1 icmp_seq=4 ttl=255 time=1.746 ms
84 bytes from 192.168.38.1 icmp_seq=5 ttl=255 time=1.979 ms

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.258 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.974 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=2.941 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.813 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.905 ms

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.38.1  4.383 ms  2.333 ms  2.456 ms
 2 18.18.11.2  4.307 ms  2.418 ms  3.309 ms
 3 *192.168.50.10  3.744 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.38.1  1.928 ms  1.638 ms  3.372 ms
 2 18.18.11.2  2.728 ms  3.551 ms  2.538 ms
 3 *192.168.50.10  5.579 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.38.1  2.662 ms  2.728 ms  4.075 ms
 2 18.18.38.2  3.423 ms  4.458 ms  5.413 ms
 3 18.18.28.1  6.538 ms  6.485 ms  2.458 ms
 4 *192.168.50.10  3.976 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS>
```

Gambar 37

4. Mengaktifkan kembali kedua link Cisco-Huawei dan memastikan jalur routing kembali ke jalur utama.

```

84 bytes from 192.168.50.10: icmp_seq=1 ttl=62 time=0.258 ms
84 bytes from 192.168.50.10: icmp_seq=2 ttl=62 time=0.970 ms
84 bytes from 192.168.50.10: icmp_seq=3 ttl=62 time=0.500 ms
84 bytes from 192.168.50.10: icmp_seq=4 ttl=62 time=0.613 ms
84 bytes from 192.168.50.10: icmp_seq=5 ttl=62 time=0.500 ms

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.30.1 0.383 ms 2.333 ms 2.456 ms
 2 10.10.10.2 4.907 ms 2.815 ms 3.359 ms
 3 *192.168.50.10 3.744 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.30.1 1.928 ms 1.638 ms 3.372 ms
 2 10.10.11.2 2.728 ms 3.551 ms 2.538 ms
 3 *192.168.50.10 5.570 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.30.1 3.662 ms 2.728 ms 0.075 ms
 2 10.10.10.2 3.053 ms 6.405 ms 5.013 ms
 3 10.10.20.1 6.538 ms 6.405 ms 2.458 ms
 4 *192.168.50.10 2.776 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.30.1 2.482 ms 2.111 ms 2.072 ms
 2 10.10.10.2 2.911 ms 1.594 ms 2.265 ms
 3 *192.168.50.10 3.728 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS>

```

Gambar 38

2 Analisis Hasil Percobaan

Pada praktikum ini seluruh konfigurasi berhasil dilakukan sesuai dengan tujuan yang telah ditentukan. Pada praktikum pertama, VLAN berhasil dibuat pada Cisco Switch dan setiap access port telah ditempatkan pada VLAN yang sesuai. Konfigurasi trunk juga berhasil membawa seluruh VLAN menuju Cisco Router.

Pada praktikum kedua, implementasi Router on a Stick berhasil dilakukan sehingga Cisco Router dapat berfungsi sebagai gateway bagi seluruh VLAN. DHCP Server pada VLAN 10 dan VLAN 20 berhasil memberikan alamat IP secara otomatis kepada client, sedangkan VLAN 30 dan VLAN 40 menggunakan konfigurasi IP statis. Seluruh host dapat melakukan komunikasi dengan gateway maupun antar VLAN.

Pada praktikum ketiga, konfigurasi alamat IP pada Huawei Router, Cisco Router, dan MikroTik berhasil dilakukan. Seluruh link antar router berada dalam kondisi aktif dan setiap router dapat berkomunikasi dengan router lain melalui pengujian ping.

Pada praktikum keempat, implementasi OSPF multivendor berhasil dilakukan. Cisco Router, Huawei Router, dan MikroTik berhasil membentuk neighbor OSPF serta saling bertukar informasi routing. Routing table pada masing-masing router menunjukkan bahwa seluruh network telah dipelajari melalui OSPF. Pengujian konektivitas antara VLAN Cisco dan LAN Huawei juga berhasil dilakukan tanpa adanya packet loss.

Pada praktikum kelima, mekanisme failover OSPF berhasil diamati. Saat kedua link utama Cisco-Huawei aktif, trafik melewati jalur utama dengan cost terendah. Ketika salah satu link dimatikan, komunikasi tetap berjalan melalui link utama yang masih aktif. Saat kedua link utama dimatikan, OSPF secara otomatis mengalihkan trafik melalui MikroTik sebagai jalur cadangan. Setelah kedua link utama diaktifkan kembali, trafik kembali menggunakan jalur utama sesuai dengan nilai cost yang telah dikonfigurasi.

Secara keseluruhan, hasil praktikum sesuai dengan teori mengenai VLAN, inter-VLAN routing, DHCP, OSPF multivendor, dan mekanisme failover pada jaringan dinamis.

3 Hasil Tugas Modul

Tugas Modul kelompok ada pada file Readme

4 Kesimpulan

Berdasarkan praktikum yang telah dilakukan, konfigurasi VLAN pada Cisco Switch berhasil diterapkan dengan baik sehingga setiap host dapat ditempatkan pada segmen jaringan yang sesuai. Implementasi Router on a Stick pada Cisco Router memungkinkan komunikasi antar VLAN berjalan dengan baik serta mendukung layanan DHCP untuk beberapa VLAN.

Konfigurasi alamat IP antar router pada Cisco, Huawei, dan MikroTik juga berhasil dilakukan sehingga seluruh perangkat dapat saling terhubung. Penerapan protokol routing OSPF multivendor berhasil membentuk hubungan neighbor dan mendistribusikan informasi routing ke seluruh jaringan.

Pengujian failover menunjukkan bahwa OSPF mampu melakukan perpindahan jalur secara otomatis ketika jalur utama mengalami gangguan dan kembali menggunakan jalur utama setelah koneksi pulih. Dengan demikian, seluruh tujuan praktikum berhasil dicapai dan hasil yang diperoleh sesuai dengan teori yang dipelajari.

5 Lampiran

5.1 Dokumentasi saat praktikum



Gambar 39: Dokumentasi Praktikan